1. What is OOPS

OOPS is a way of organizing code that uses object and classes to represent real-world entities and their behaviour. In OOP, object has attributes thing that has specific data and can perform certain actions using methods.

2. Components in Python

* Function
* Object and class
* Modules
* Build in Functions

3. Access a class variable of city = 'Chennai' and pass ~~name~~ same value for all objects

```
class Student:
    def __init__(self, name):
        self.name = name

    def dis(self):
        print(self.name)

s1 = Student('prem')
s1.dis()
s2 = Student('Fin')
s2.dis()

s3 = Student('Allen')
s3.dis()
```

```
s4 = Student('Bim')
s4.dis()
s5 = Student('Hin')
s5 = dis()
```

5. Create a class with instance variable of name, age, city and four instance method for the same with 5 object creation

```
class Form:
    def __init__(self, name, age, city):
        self.name = name
        self.age = name
        self.city = name

    def dis(self):
        print(self.name)
        print(self.age)
        print(self.city)
h1 = Form('Prem', 22, 'chennai')
h1.dis()
h2 = Form('arun', 23, 'erode')
h3 = Form('Fin', 20, 'NYC')
h4 = Form('Sao, 20, 'bgr')
h5 = Form('dan, 24, 'chennai')
```

6. Types of
Single inheri
Multiple inher
Multilevel i
Hierarchical
Hybrid i

7. Scenario
Single inheri
```
class Person
    def dis
        print
class Employ
    def sal
        pri
```
Multiple inh
```
class St
    def
        p
class Te
    def
class T
    p
te=Teach
```

6. Types of inheritance

Single inheritance

Multiple inheritance

Multilevel inheritance

Hierarchical Inheritance

Hybrid inheritance

7. Scenario of inheritance

Single inheritance

```
class person:
    def display(self):
        print('person details')
class Employee(Person):
    def salary(self):
        print('Employee salary')
```

Multiple inheritance:

```
class Student:
    def study(self):
        print('studying')
class Teacher:
    def tech(self):
        print('Teaching')
class TeachingAssistance(Student, Teacher):
        pass
ta=TeachingAssistance()
ta.study()
ta.tech()
```

use of __init__

__init__ is a Constructor in Python. which is automatically called when a new object of a class is Created. Its main purpose is to initialize the object's attributes and set up its initial state.